

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF NATURAL SCIENCES

Term: **Quiz 1 (SET 1)**Course Number: **Math 4241**

Course Title: Integral Calculus and Differential Equations

Full Marks: 15

Time: 30 Minutes

Answer all the questions. The symbols have their usual meanings. Marks of each question are written in the brackets.

1. a)	Define inscribed rectangle. Find the area of the region bounded by the graph of $f(x) = \sqrt{x}$, and the x -axis for $0 \leq x \leq 1$. Use the method for Partition with Subintervals of Unequal Widths and right endpoint approximation.	(1+5)
b)	Define ODE. Using the Reduction Formula evaluate the definite Integral for $\sin^5 x$ within the interval $0 \leq x \leq \frac{\pi}{2}$.	(1+4)
c)	Define Definite integral.	(2)
d)	Solve the initial value problem: $\frac{dy}{dx} + 3y = xy^2$; $y(0) = 1$.	(5)
e)	Write down the order and degree of the following differential equations, also mention if the differential equation is linear or non-linear. (i) $\left(\frac{d^3y}{dx^3}\right)^2 + x \frac{d^2y}{dx^2} = x^2$ (ii) $\frac{d^3y}{dx^3} + x \frac{d^2y}{dx^2} + x \frac{dy}{dx} + 2y = e^x$	(1+1)

Name:**Student ID:**

(Start writing from the line below)